

were no boundary lines or borders of countries," he said. "The second was when I saw India. It looked very big and grand, not like the 2D paper maps of it." He would spend 18 days on the ISS and carry out experiments proposed by Indian scientists from R&D labs across the country on the effects of microgravity, a state of lightness that allows astronauts and objects to float about. He would study its effects on (highly nutritious) microalgae, food crop seeds and metabolic supplements. Microgravity degenerates human muscle—Shukla would lose 4.2 kilos of weight over the course of the mission—and astronauts are required to exercise vigorously daily. He would also gain a few centimetres in height due to the elongation of his spine—a common microgravity effect—but that would be reversed back on Earth.

"When I first saw India from orbit, it looked very big and grand," says Shukla. "It wasn't like the 2D maps of it"

Shukla's trip was a stepping stone for India because "with a human space mission, you learn things you would otherwise not know... You're launching a space capsule with human lives on board. It changes the scale and the magnitude of work that is involved," he said. In his three interactions with children from the ISS through video-conferencing and ham radio, the one query that was most popular was, "How do I become an astronaut?" That, according to the Indian astronaut, was "the biggest win of this mission". ■